Network Science

Lecture 07: Small-World Networks

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

Speaker notes

This lecture asks two deceptively similar questions that turn out to be very different: (1) Why do short paths exist in large networks? (2) Can people *find* those short paths using only local information? The Watts-Strogatz model answers the first; Kleinberg's grid model answers the second. The gap between the two is the navigational gap — the fifth kind of gap in the course arc. Milgram's letter-chain experiment (1967) is the running example that threads the entire lecture.

Milgram's Experiment — The Question That Started It All

In 1967, the social psychologist Stanley Milgram mailed 296 letters to random people in Nebraska and Kansas. Each letter named a **target person** — a stockbroker in Boston — and asked the recipient to forward it to someone they knew *on a first-name basis* who might be "closer" to the target.

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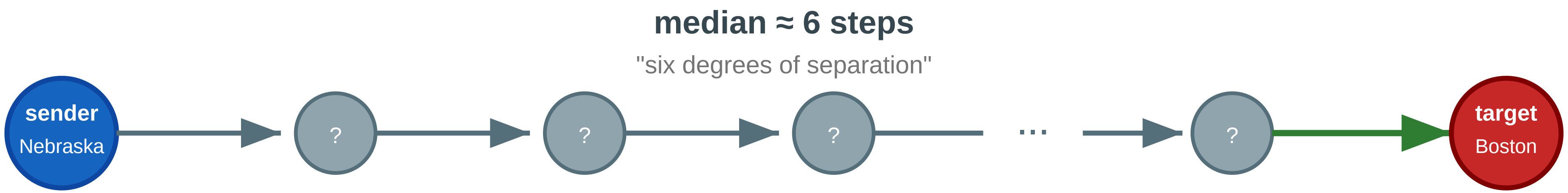
The letters that arrived took a **median of about 6 steps**.

The phrase "six degrees of separation" entered popular culture — but the experiment raised more questions than it answered.

The Chain

Milgram's small-world experiment (1967)

Can a letter reach a stranger in Boston through a chain of personal acquaintances?



Questions this experiment raises

① Why do short paths exist at all in a network of 200 million people?

② How can a network be locally clustered AND have short paths to strangers?

③ If short paths exist, can ordinary people find them using only local knowledge?

④ Only ~25% of chains arrived. Why did most fail?

> Each intermediary sees only their own contacts and must guess who is "closer" to the target. About 25% of chains completed; the rest died when someone refused to forward. The experiment demonstrates both that short paths *exist* and that finding them is *hard*.

Speaker notes

This figure is the lecture anchor. Question 1 (why short paths?) leads to Module 1. Question 2 (clustering + short paths?) leads to Watts-Strogatz. Question 3 (can people find them?) leads to Kleinberg. Question 4 (why do most chains fail?) leads to the empirical anchor (Dodds et al. 2003). Refer back throughout.

Four Questions from One Experiment

1. **Existence.** Why do short paths exist at all in a network of 200 million people?
2. **Coexistence.** How can a network be *locally clustered* (your friends know each other) and *globally short* (six steps to a stranger in another state)?
3. **Navigability.** If short paths exist, can ordinary people find them using only local knowledge — no map, no search engine?
4. **Failure.** Only ~25% of chains arrived. Does this mean short paths are rare, or that *finding* them is the bottleneck?

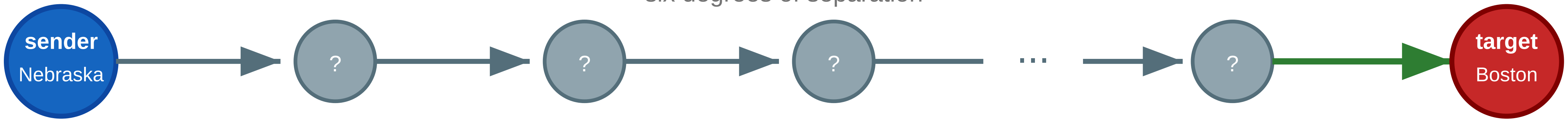
Every concept in today's lecture answers one of these.

Milgram's small-world experiment (1967)

Can a letter reach a stranger in Boston through a chain of personal acquaintances?

median ≈ 6 steps

"six degrees of separation"



Questions this experiment raises

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③ If short paths exist, can ordinary people find them using only local knowledge?

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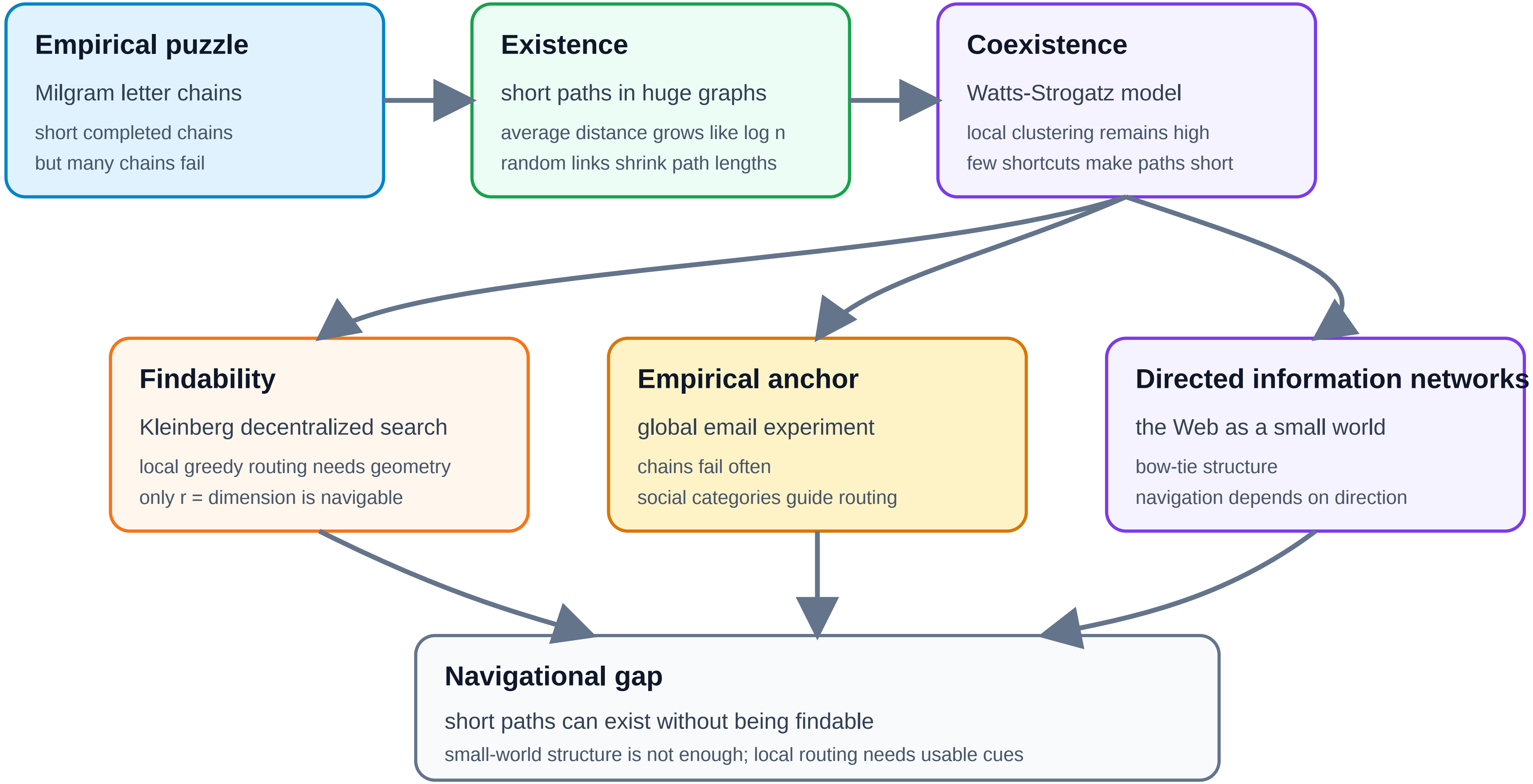
Speaker notes

Questions 1–2 map to Modules 1–2 (small-world phenomenon, Watts-Strogatz). Question 3 maps to Module 3 (Kleinberg). Question 4 maps to Module 4 (Dodds et al. empirical test). The Web module (Module 5) extends the navigability question to directed information networks.

Argument Map



Speaker notes

This map separates the lecture's two central questions. First, why do short paths exist in large networks? Logarithmic distances and Watts-Strogatz shortcuts answer that. Second, can actors find those paths with only local information? Kleinberg shows that findability requires the right geometry of long-range links, and the empirical email experiment shows that many chains fail even when short paths probably exist. The Web module transfers the same navigability issue to directed information networks.

A New Kind of Gap — Existence vs. Findability

L03–L04 had a **computational** gap (NP-hard ideals). L05 had a **causal** gap. L06 had a **structural** gap (completeness required).

L07 has a **navigational** gap: short paths may *exist* without anyone being able to *find* them using local information.

Lecture	Gap type	Ideal	Reality	Strategy
L03–L04	Computational	Exact optimization	NP-hard	Polynomial heuristics
L05	Causal	Mechanism identification	Snapshots insufficient	Longitudinal data
L06	Structural	Complete signed graph	Sparse, noisy data	Approximate balance
L07	Navigational	Short navigable paths	Local information only	Geometric link distribution

Speaker notes

The navigational gap is structurally different from the previous ones: it is not about what we can *compute* or *measure*, but about what *actors inside the network* can do with limited information. The fix is not a better algorithm run by an external analyst — it is a specific *geometry* of long-range links that makes local decisions globally effective. Kleinberg's result identifies this geometry precisely.

Takeaway

> The small-world phenomenon is really two phenomena: the *existence* of short paths (explained by a few random long-range links) and the *findability* of those paths (explained only by a specific geometric distribution of links). Milgram's experiment revealed both — and the gap between them.

The Small-World Phenomenon

> **Guiding Question:** Why are distances in large networks often surprisingly short?

Speaker notes

This module establishes the basic empirical fact — typical distances grow logarithmically with network size — and gives examples from social, collaboration, and communication networks.

Logarithmic Distances

Small-world property. A network exhibits the small-world property if the average shortest-path distance $\bar{d}$ grows at most logarithmically with the number of nodes:

$$\bar{d} \propto \log |V|$$

Recall: $d_G(u, v)$ is the **graph distance** between two nodes — the number of edges in a shortest path. The average shortest-path distance is the graph-level average over node pairs: $\bar{d} = \frac{1}{|V|(|V|-1)} \sum_{u \neq v} d_G(u, v)$

In a random graph $G(n, p)$ with average degree k, the typical distance is approximately $\bar{d} \approx \frac{\log n}{\log k}$.

Speaker notes

The logarithmic scaling is the formal content of "six degrees of separation." It explains why Milgram's result is not mysterious from a graph-theory perspective: in any network where each node has tens or hundreds of contacts, the branching factor is so high that $\log n / \log k$ remains small even for billions of nodes. The surprise is not that short paths exist — it is that ordinary people can sometimes find them.

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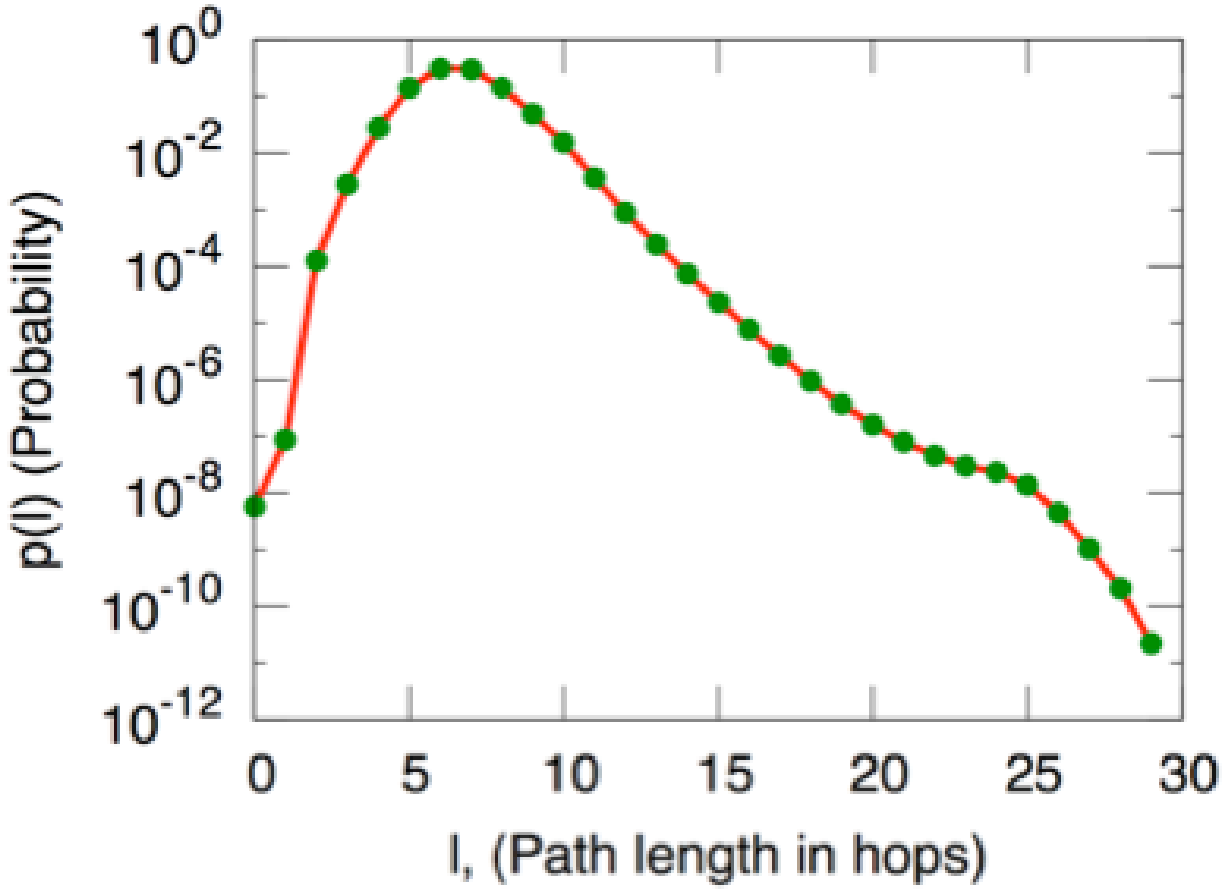
For a social network of $n = 300$ million people with average degree $k \approx 100$: $\bar{d} \approx \frac{\log(3 \times 10^8)}{\log 100} \approx \frac{19.5}{4.6} \approx 4.2$. Milgram's 6 is in the right ballpark.

Empirical Examples



Source: Easley & Kleinberg 2010, p. Ch. 20

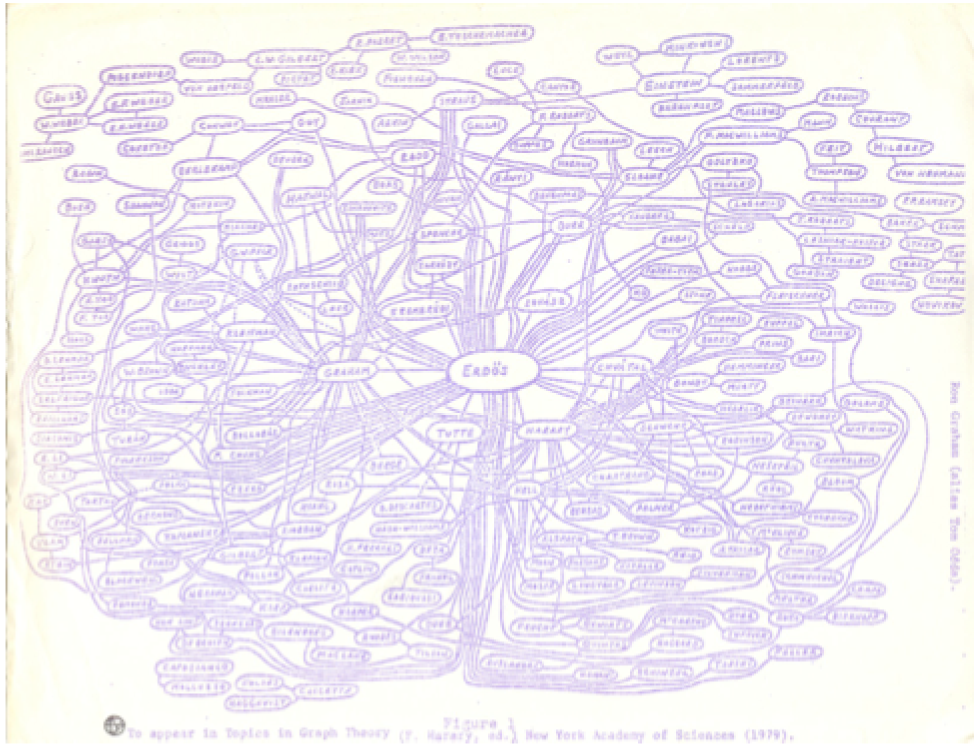


Figure 2.12: Ron Graham's hand-drawn picture of a part of the mathematics collaboration graph, centered on Paul Erdös [189]. (Image from http://www.oakland.edu/enp/cgraph.jpg)

Source: Easley & Kleinberg 2010

Left: Microsoft Instant Messenger network — 180 million users, median distance ~7. **Right:** Mathematical collaboration network — Erdős numbers rarely exceed 5. Both confirm logarithmic scaling across orders of

Speaker notes

The IM data (Leskovec & Horvitz 2008) is especially striking: 180 million nodes, 1.3 billion edges, average distance 6.6. The Erdős collaboration graph is smaller (~400k mathematicians) but culturally iconic. Both examples reinforce that logarithmic distances are a robust empirical fact, not a quirk of Milgram's Nebraska sample.

Takeaway

> Logarithmic distances are a robust empirical fact across social, communication, and collaboration networks. In any network with moderate average degree, even billions of nodes remain only a handful of hops apart.

Quick Check

The Facebook social graph has approximately 3 billion users and an average degree of about 300.

1. Estimate the average shortest-path distance using $\bar{d} \approx \log n / \log k$.
2. Facebook reported an empirical average distance of 3.57 in 2016. Is this consistent with your estimate?
3. Why might the empirical value be *lower* than the logarithmic estimate?

Speaker notes

Give them 3 minutes. (1) $\bar{d} \approx \log(3 \times 10^9)/\log(300) \approx 21.8/5.7 \approx 3.8$. (2) Yes — 3.57 is close to 3.8. (3) The formula assumes a random graph; Facebook has hubs (celebrities, organizations) with degrees far above 300, which create even shorter paths. The degree distribution is heavy-tailed, not Poisson, so a few high-degree nodes pull the average distance below the random-graph prediction.

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Quick Check — Solution

1. $\bar{d} \approx \frac{\log(3 \times 10^9)}{\log 300} \approx \frac{21.8}{5.7} \approx 3.8$

2. Yes — 3.57 is remarkably close to the logarithmic estimate.

3. The $\log n / \log k$ formula assumes homogeneous degree (random graph). Facebook has heavy-tailed degree — celebrities and organizations with millions of friends act as hubs, creating shortcuts that reduce $\bar{d}$ below the random-graph prediction.

The Watts–Strogatz Model

> **Guiding Question:** How can a network be both highly clustered *and* have short paths?

Speaker notes

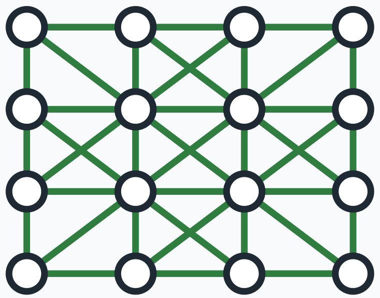
This is the paradox that Watts & Strogatz (1998) resolved. Regular lattices have high clustering but long paths. Random graphs have short paths but no clustering. Real social networks have *both*. The Watts-Strogatz model shows that a tiny amount of randomness bridges the gap.

The Paradox

Real social networks have two seemingly contradictory properties:

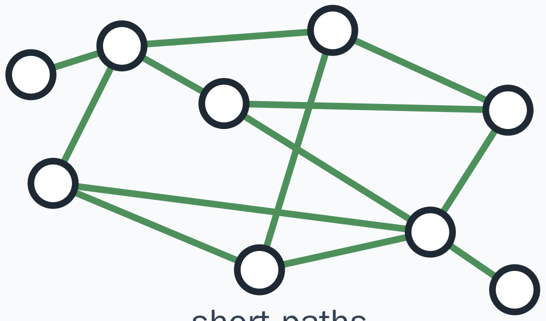
Property	Regular lattice	Random graph	Real social network
Clustering C	High (~ 0.5)	Low ($\sim k/n$)	**High** (~ 0.1–0.5)
Avg. path length L	Long ($\sim n/2k$)	Short ($\sim \log n/\log k$)	**Short** ($\sim \log n$)

Regular lattice



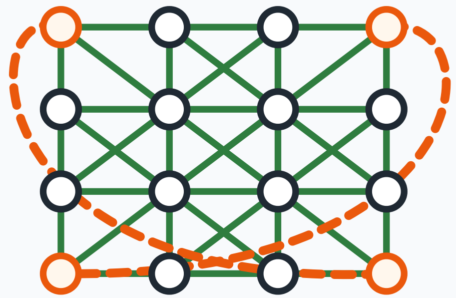
local, regular structure
high clustering, long paths

Random graph



short paths
but low clustering

Small-world graph



local lattice + shortcuts
high clustering, short paths

Speaker notes

The table makes the paradox concrete. The challenge is to find a model that produces the bottom-right cell: high clustering AND short paths. Neither a lattice nor a random graph does this alone. The Watts-Strogatz model interpolates between the two extremes with a single parameter p.

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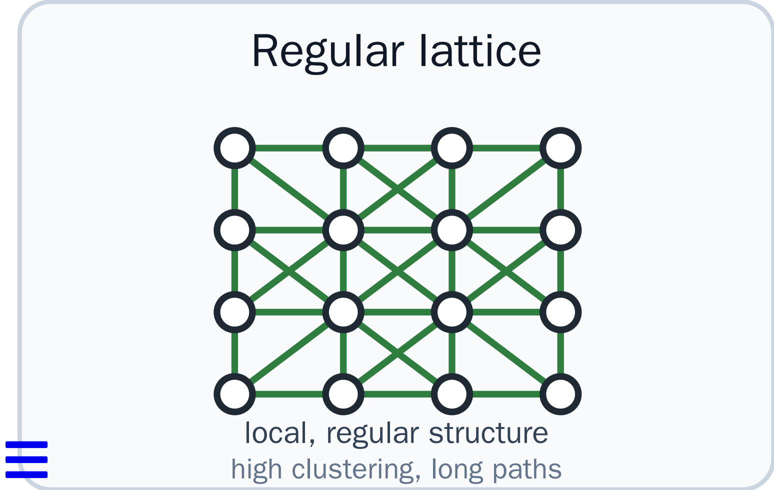
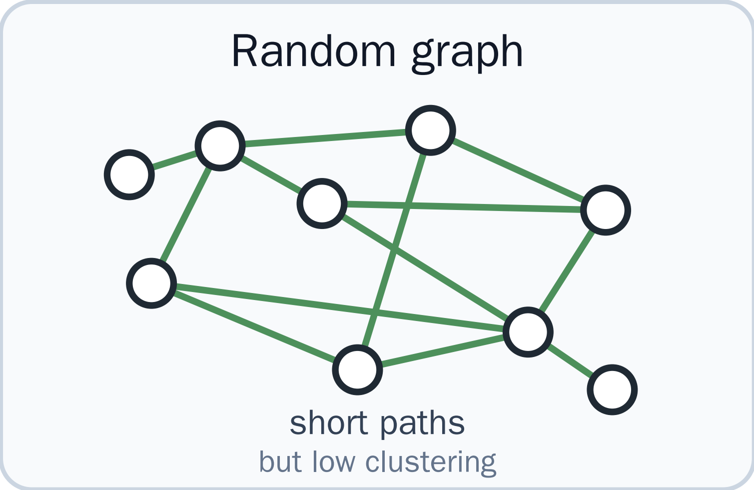
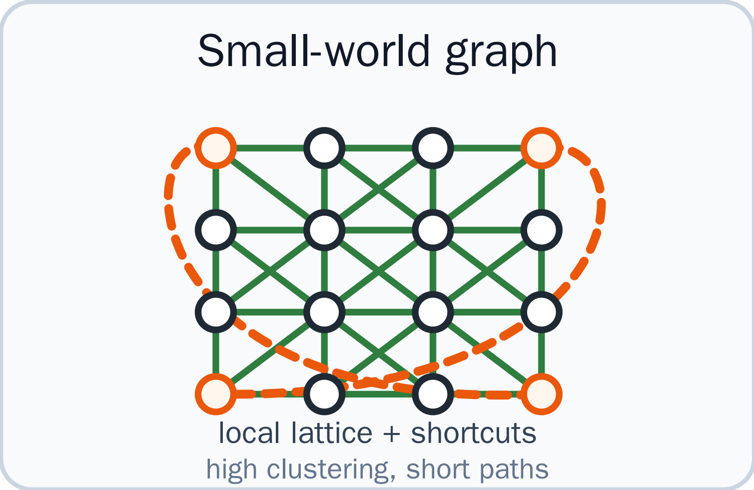
Avg. path length L	Long ($\sim n/2k$)	Short ($\sim \log n/\log k$)	**Short** ($\sim \log n$)

A regular lattice is clustered but large. A random graph is small but unclustered. **Real networks are both** — high C and low L simultaneously.







The Model

Watts–Strogatz model (1998).

1. Start with a **regular lattice**: n nodes arranged in a regular local structure, each connected to nearby neighbors.
2. For each edge, with probability p, **rewire** one endpoint to a uniformly random node (avoiding self-loops and duplicates).

The parameter $p \in [0, 1]$ controls the interpolation:

- $p = 0$: regular lattice (high C, high L)
- $p = 1$: random graph (low C, low L)
- $0 < p \ll 1$: **small-world regime** (high C, low L)

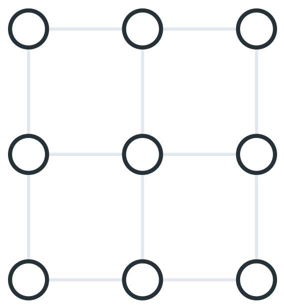
Speaker notes

The model is beautifully simple: one parameter, one structural transition. The key insight is that the transition in $L(p)$ and $C(p)$ happens at *different* speeds. Path length collapses rapidly (a few random shortcuts suffice) while clustering decays slowly (most triangles survive). This creates a wide window where $C(p) \approx C(0)$ but $L(p) \approx L(1)$ — the small-world regime.

Watts–Strogatz in Four Steps

1. Place nodes

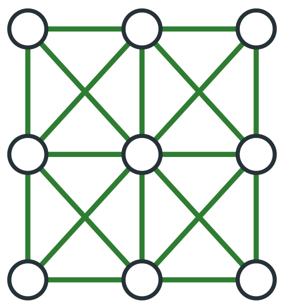
arrange n nodes on a lattice



positions are regular

2. Connect neighbors

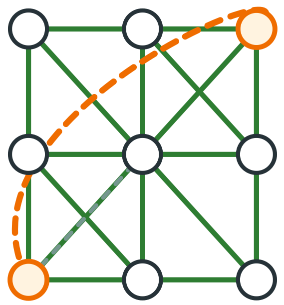
local edges create clustering



high C, long L

3. Rewire with p

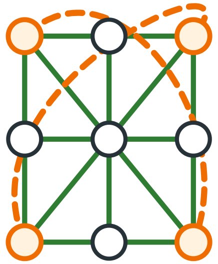
move a few endpoints randomly



small p creates shortcuts

4. Result

small-world regime

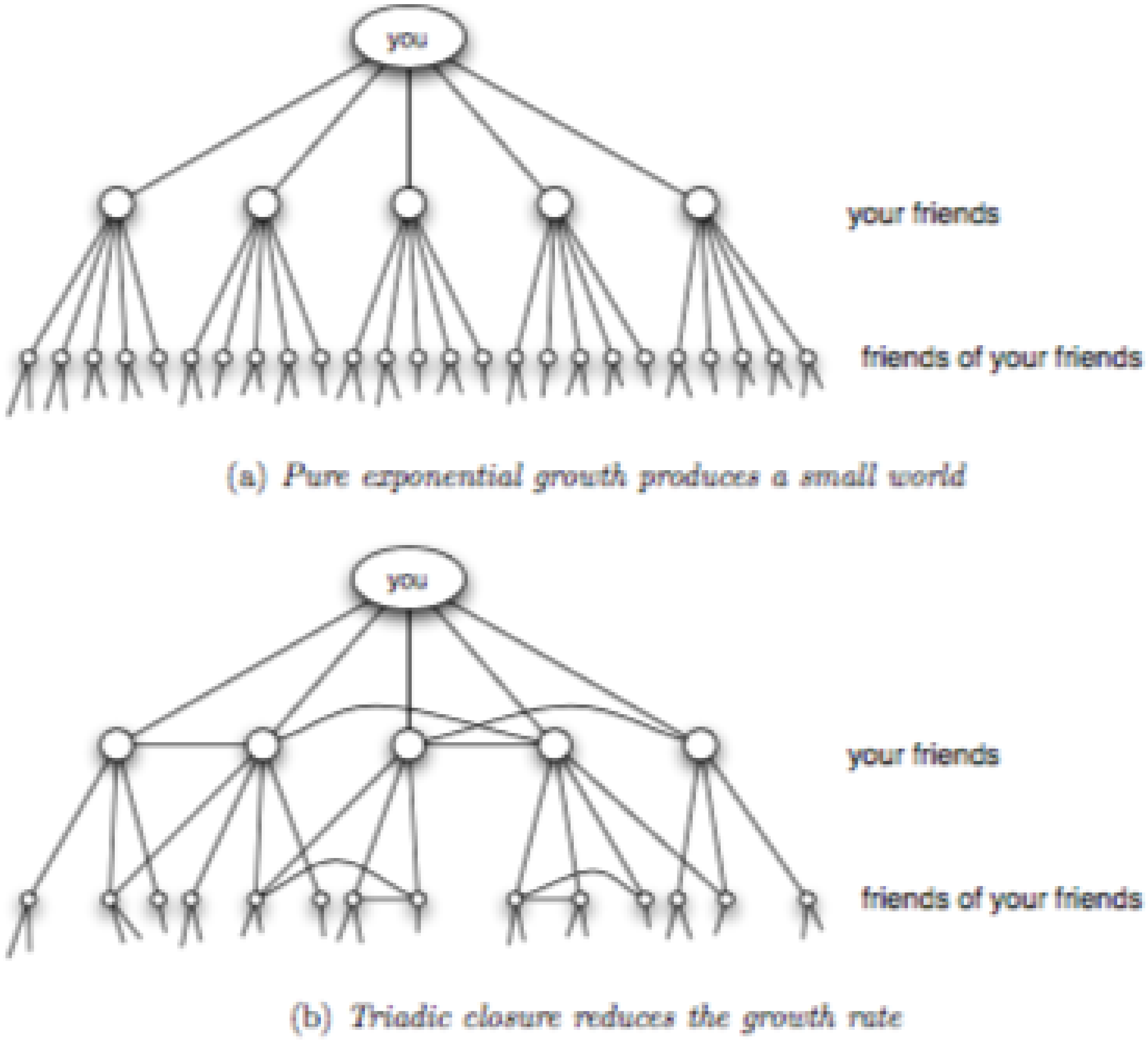


high C
short L

Speaker notes

Use this as the visual bridge from the definition to the characteristic curves. Step 1 places nodes on a regular lattice. Step 2 adds local edges, which preserve clustering but have long paths. Step 3 rewires a few edge endpoints with probability p, producing long-range shortcuts. Step 4 is the small-world regime: most local structure survives, but the shortcuts collapse path length.

Why Clustering Makes Paths Longer



Source: Easley & Kleinberg 2010, p. Ch. 20

If every friend led to entirely new people, reachability would grow like a tree and paths would be very short. In real friendship networks, friends of friends often overlap, creating triangles and high clustering. That overlap slows the expansion of the search frontier.

Speaker notes

Use this figure to separate two mechanisms. The top panel is the branching-process intuition behind short paths: each step reaches many new people. The bottom panel is what triadic closure does: your friends often know the same people, so the second neighborhood is smaller than a tree would predict. The Watts-Strogatz idea is to preserve this local overlap while adding a few long-range shortcuts that restore fast global reach.

Takeaway

The Watts-Strogatz model resolves the clustering/path-length paradox: a tiny fraction of random long-range shortcuts collapses distances without destroying local structure. The small-world regime exists because $L(p)$ falls much faster than $C(p)$.

Quick Check

A regular local lattice has $n = 1000$ nodes and $k = 10$ neighbors per node.

1. At $p = 0$, should the average path length L be closer to a lattice scale or a random-graph scale?
2. After rewiring 1% of edges ($p = 0.01$), how many long-range shortcuts are created?
3. Qualitatively, what happens to L and C?

Speaker notes

Give them 3 minutes. (1) At $p = 0$, path length is governed by the lattice geometry: movement is local, so L is large compared with a random graph of the same size and average degree. Avoid the old ring-lattice estimate here; the visual model is now a regular local lattice. (2) The lattice has $nk/2 = 5000$ edges; 1% rewired = 50 long-range shortcuts. (3) L drops dramatically because shortcuts cross the lattice, while C drops only slightly because most local edges remain.

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Quick Check — Solution

1. Closer to a **lattice scale**: paths are still local and grow with the lattice diameter, not with $\log n$.

2. $nk/2 = 5000$ edges total; 1% = **50 long-range shortcuts**.

3. L drops sharply because shortcuts jump across the lattice. C barely changes because only 50 of 5000 edges were rewired, so the vast majority of local neighborhoods survive. **This is the small-world regime.**

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Kleinberg's Decentralized Search

> **Guiding Question:** Short paths exist — but can local actors *find* them without a map?

Speaker notes

This is where the navigational gap opens. Watts-Strogatz proved that short paths exist; Kleinberg (2000) asked whether a *decentralized* agent — one that sees only its own neighbors and knows the target's approximate location — can find them efficiently. The answer is: only if the long-range links have a very specific geometric distribution.

The Problem

In Milgram's experiment, each person forwarded the letter based on **local information only**:

- they knew their own contacts
- they knew the target's name, city, and profession
- they did *not* know the global network structure

Speaker notes

This is the conceptual crux of the lecture. Existence of short paths is a property of the graph. Findability of short paths is a property of the *algorithm plus the graph*. Watts-Strogatz gives existence; Kleinberg gives the conditions for findability.

The Problem

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Decentralized search: at each step, the current holder forwards to the neighbor *closest* to the target (in some distance metric). No global routing table, no BFS.

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Decentralized search: at each step, the current holder forwards to the neighbor *closest* to the target (in some distance metric). No global routing table, no BFS.

Watts-Strogatz does not guarantee this works. Short paths exist, but the greedy algorithm may not find them — it depends on *where* the long-range links point.

Kleinberg's Grid Model (2000)

Setup. Place n nodes on a d-dimensional grid. Each node has:

- **local contacts:** all grid neighbors within lattice distance p *(the local radius; in 2D, usually Manhattan distance* $|x_u - x_v| + |y_u - y_v|$*)*
- **one long-range link:** drawn with probability proportional to $d(v, u)^{-r}$, where d is grid distance and $r \geq 0$ is the distance exponent.

Greedy routing. At each step, forward to the neighbor (local or long-range) closest to the target in grid distance.

The parameter r controls the *geometry* of long-range links:

- $r = 0$: uniform random (long-range links go anywhere)
- $r = d$: inverse-power law matched to dimension (links span all scales equally)
- $r \gg d$: long-range links are effectively local

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Speaker notes

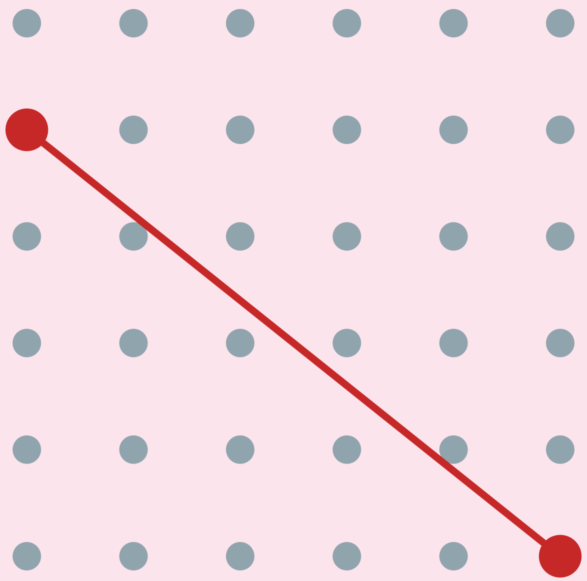
The model separates the two ingredients cleanly: the grid provides the local clustering, and the long-range links provide the shortcuts. The exponent r is the single knob that controls navigability. Kleinberg's result is that only one value of r makes greedy search efficient.

The Three Regimes

Kleinberg's grid model — the geometry of navigability

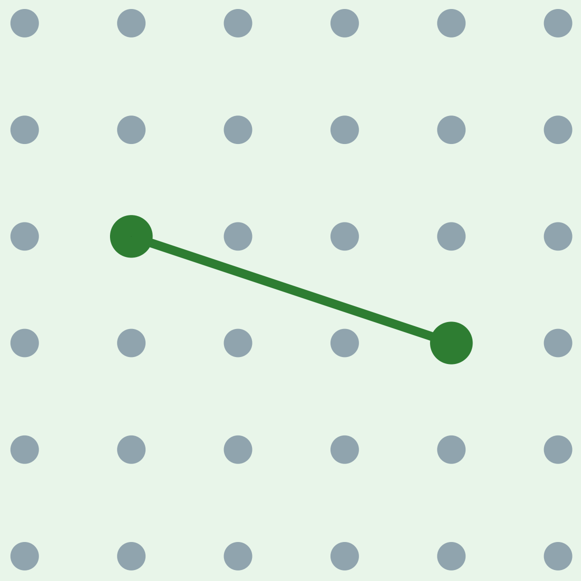
Each node has local grid neighbors + one long-range link drawn with $P(\text{link to } u) \propto d(v,u)^{-r}$

$r = 0$ (uniform random)



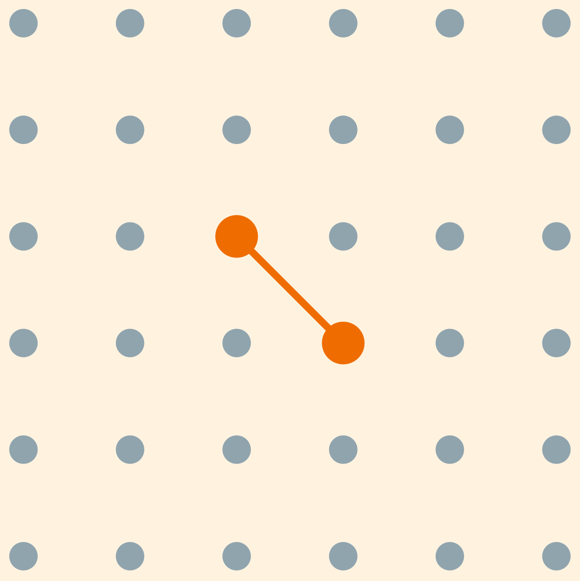
short paths exist,
but search is $\Omega(n^{2/3})$

$r = 2 = d$ (inverse-square)



short paths exist
AND search is $O(\log^2 n)$

$r \gg d$ (too clustered)



short paths exist,
but "shortcuts" are too local

Speaker notes

The figure shows why $r = d$ is special. At $r = 0$, long-range links are uniformly random — they create short paths but provide no geographic gradient for the greedy algorithm to follow. At $r \gg d$, links are so local that they add no useful shortcuts. Only at $r = d$ do the links span all scales proportionally, creating a hierarchy of shortcuts that the greedy algorithm can exploit at every distance.

Kleinberg's Theorem

Theorem (Kleinberg, 2000).

In the d-dimensional grid model with long-range exponent r:

- If $r = d$: greedy routing finds the target in $O(\log^2 n)$ steps.
- If $r \neq d$: decentralized search has a **polynomial lower bound**: at least $\Omega(n^c)$ steps for some fixed constant $c = c(r, d) > 0$.

Speaker notes

This is one of the sharpest results in the course. The navigability transition is not gradual — it is a threshold phenomenon. At $r = d$, greedy search succeeds in $O(\log^2 n)$ steps; at any other r, the lower bound is polynomial in n: $\Omega(n^c)$ for some fixed $c > 0$ depending on the model parameters. Emphasize the asymptotic point: for finite classroom-sized n, $n^{0.001}$ can look small, but asymptotically every fixed positive power dominates every polylogarithm. The uniqueness of $r = d$ means that navigability is not a generic property of small-world networks — it requires a specific geometry that most random-shortcut models (including Watts-Strogatz) do not guarantee.

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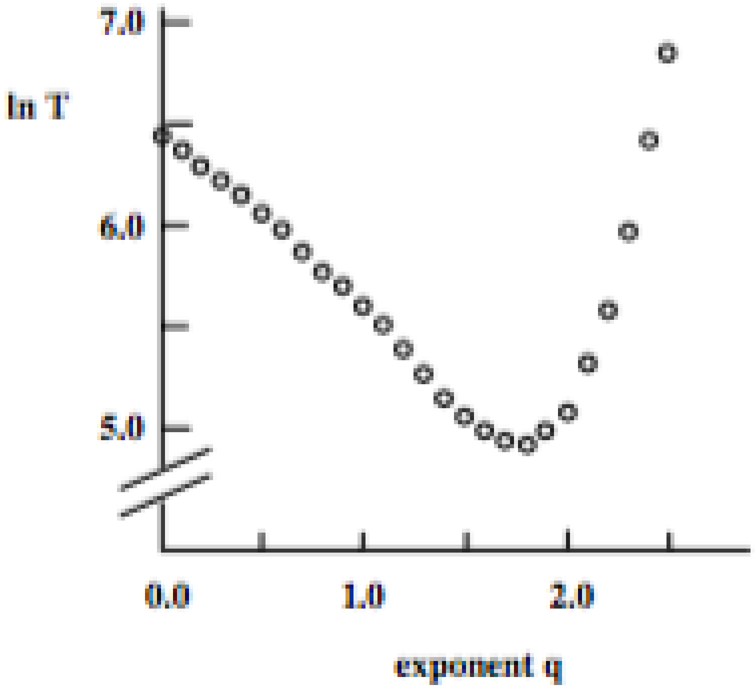
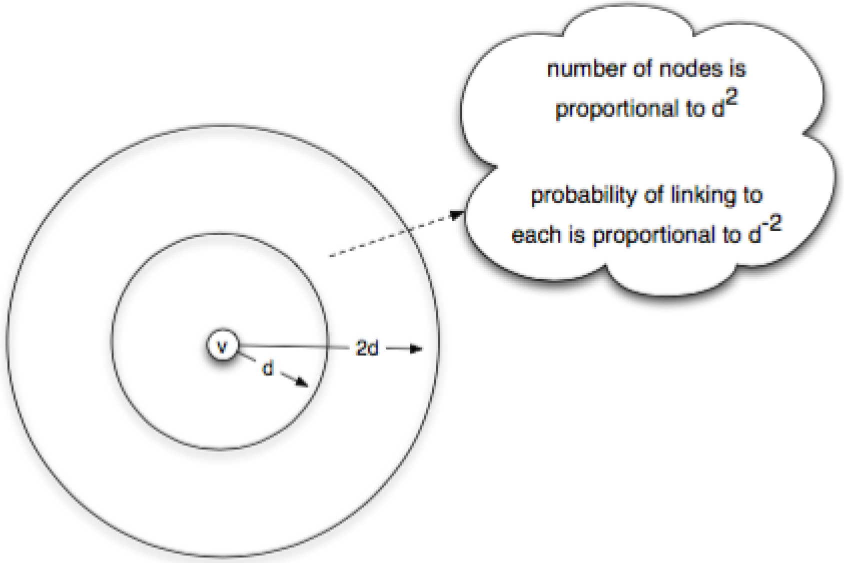
- If $r = d$: greedy routing finds the target in $O(\log^2 n)$ steps.
- If $r \neq d$: decentralized search has a **polynomial lower bound**: at least $\Omega(n^c)$ steps for some fixed constant $c = c(r, d) > 0$.

The exponent $r = d$ is **unique** — it is the only value at which decentralized search is polylogarithmic. Any fixed positive power n^c eventually grows faster than $\log^2 n$, even if c is small.

Simulation Evidence



Source: Easley & Kleinberg 2010, p. Ch. 20



Source: Easley & Kleinberg 2010, p. Ch. 20

Left: delivery time vs. exponent r — a sharp minimum at $r = d = 2$. **Right:** the proof intuition — at $r = d$, each "ring" of nodes at distance 2^i from the target receives roughly equal link probability, creating a hierarchy of scales the algorithm descends one level at a time.

Speaker notes

The simulation plot is the visual proof that the theorem is not just asymptotic: even on moderate-sized grids (hundreds to thousands of nodes), the minimum at $r = d$ is sharp and clearly visible. The "equal probability per ring" intuition is the heart of the proof: if each doubling of distance carries equal total link probability, then a greedy step reduces the remaining distance by a constant factor with constant probability, yielding $O(\log n)$ rounds of $O(\log n)$ steps each.

Takeaway

Short paths exist in almost any small-world network. But navigability — the ability of local actors to *find* those paths — requires a specific geometric distribution of long-range links ($r = d$). Watts-Strogatz gives existence; Kleinberg gives findability.

Quick Check

A social network is well-modeled by a 2D grid with random long-range links.

1. What exponent r makes greedy search efficient?
2. If the network's long-range links are uniformly random ($r = 0$), what happens to (a) the existence of short paths and (b) greedy routing performance?
3. Why is the Watts-Strogatz model insufficient to explain Milgram's result?

Speaker notes

Give them 4 minutes. (1) $r = 2$ (matches the grid dimension $d = 2$). (2a) Short paths still exist — random shortcuts reduce L. (2b) Greedy routing degrades to polynomial time — the shortcuts point to random locations that provide no useful gradient toward the target. (3) Watts-Strogatz shows short paths *exist* but uses uniform rewiring ($r = 0$), which does not support efficient decentralized search. Milgram's participants actually *found* short paths, which requires $r = d$ — a stronger condition than Watts-Strogatz guarantees.

Quick Check

A social network is well-modeled by a 2D grid with random long-range links.

1. What exponent r makes greedy search efficient?
2. If the network's long-range links are uniformly random ($r = 0$), what happens to (a) the existence of short paths and (b) greedy routing performance?
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Quick Check — Solution

1. $r = 2 = d$ for a 2D grid.

2.

- (a) Short paths **still exist** — uniform random shortcuts reduce average path length.
- (b) Greedy routing **fails** — it takes $\Omega(n^{2/3})$ steps because the random shortcuts provide no gradient toward the target.

3. Watts-Strogatz uses **uniform** rewiring ($r = 0$). This gives short paths but not *navigable* short paths. Milgram's participants actually found short paths using local information — that requires $r = d$, a much stronger structural condition.

Beyond the Grid — Where Else Does This Show Up?

Kleinberg's grid is a toy model. But the *principle* — mix **short-range** links (for local refinement) with **long-range** links at every scale (for jumping) — generalises far beyond social networks.

Kleinberg: nodes live on a known grid. Greedy routing works when links exist at multiple scales.

Vector search: documents or images live in an embedding space. The "distance" is learned from data, not given by a map.

HNSW intuition: build a graph over vectors with two kinds of movement:

- sparse upper layers for **large jumps**
- dense lower layers for **local refinement**

Search starts coarse, then descends and refines greedily near the query.

≡

Speaker notes

Bridge slide. The students have just proved that greedy routing works at $r = d$. The same broad design principle — multi-scale links plus greedy local decisions — reappears in approximate nearest-neighbour search. Keep the HNSW explanation intuitive: it is a graph index over embedding vectors. Upper layers are sparse and help make large jumps; lower layers are denser and support local refinement. Do not claim that HNSW is simply Kleinberg with $r = d$ or that the theorem directly proves HNSW complexity. The point is conceptual transfer: navigability requires a graph whose links give greedy search useful progress at multiple scales.

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Search starts coarse, then descends and refines greedily near the query.

> HNSW is not literally Kleinberg's grid. The useful analogy is the **multi-scale search structure**: jump far first, then use local links to finish.

Why This Matters for RAG and LLM Retrieval

Every retrieval-augmented generation (RAG) pipeline has the same structure:

1. **Embed** each document chunk into $\mathbb{R}^d$ (typically $d = 384$–4096).
2. **Index** the vectors in an HNSW (or IVF, or DiskANN) graph.
3. At query time, **greedy-search** from the entry point for the top-k nearest neighbours.

Speaker notes

This is the most important forward-looking slide in the lecture. Students should leave understanding that the theoretical result from 2000 foreshadows a design principle used in modern retrieval systems: build graph structure that makes greedy progress possible. Avoid overclaiming: HNSW is a practical ANN data structure, not a direct implementation of Kleinberg's theorem. Forward-reference L09, which introduces the embedding side of the story — how the vectors that live in this index are produced in the first place. The complete picture (embedding + index) requires both lectures.

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> We will come back to **how** these embeddings are constructed in **Lecture 09** (node and document representations, spectral methods, DeepWalk, node2vec, GNNs). The *geometry* of the embedding space determines whether navigation works — which closes the loop to Kleinberg.

Empirical Anchor: The Global Email Experiment

> **Guiding Question:** Does Milgram's result replicate at scale — and what does chain failure tell us?

Speaker notes

This module provides the empirical anchor for L07, following the pattern of Kossinets-Watts (L03), Zachary (L04), Christakis-Fowler (L05), and Leskovec et al. (L06). The study is Dodds, Muhamad & Watts (2003), the largest replication of Milgram's experiment.

The Study

Dodds, Muhamad & Watts (2003), *An Experimental Study of Search in Global Social Networks*, Science 301, 827–829.

They replicated Milgram's experiment using **email** — a global medium with no geographic constraint.

Parameter	Value
Participants recruited	**~60 000** from 166 countries
Target persons	**18** in 13 countries
Chains started	**24 163**
Chains completed	**384** (~1.6%)
Median chain length (completed)	**4–7 steps**

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Speaker notes

The scale is two orders of magnitude larger than Milgram's 296 letters, and the medium (email) removes geographic friction. The completed chains confirm logarithmic distances (~5 steps median). But the headline number is the *completion rate*: only 1.6% of chains reached their target. This is even lower than Milgram's ~25%, despite the ease of forwarding email.

What They Found

1. Completed chains are short. Median 4–7 steps, consistent with Milgram and with $\log n / \log k$ scaling.

2. Most chains die early. The completion rate of ~1.6% is not because paths don't exist — it's because intermediaries *choose not to forward*. The bottleneck is **motivation**, not topology.

Speaker notes

Finding 3 is the empirical bridge to Kleinberg: successful Milgram-style search is not random — it exploits social coordinates (geography, profession, organizational hierarchy) that function as approximate distance metrics. The chains that succeeded were the ones where each forwarder made a good *greedy* decision relative to the target's known attributes. This is exactly what Kleinberg's theorem predicts: navigability requires that long-range links are distributed across scales in a way that local greedy decisions can exploit.

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3. Successful chains use geography and profession as search dimensions. Forwarders who brought the letter closer in *geographic* or *professional* distance were more effective — consistent with Kleinberg's model where search exploits multiple distance dimensions simultaneously.

What the Study Could Not See

1. Chain death ≠ path absence. A chain that dies after 3 steps does not mean no short path existed — it means someone chose not to forward. The study measures *willingness to search*, not *path existence*. The true completion rate under full cooperation is unknown.

2. Selection bias in completions. The 384 completed chains are a biased sample — they may systematically over-represent easy targets (well-known, geographically close, professionally distinctive). The median chain length of 4–7 steps may underestimate the true difficulty of reaching a random target.

3. Email ≠ acquaintance. Milgram's experiment required *personal acquaintance*; the email study allowed forwarding to anyone whose email address you knew. The social network traversed by email may be larger and weaker-tied than the face-to-face network Milgram assumed.

4. No ground-truth network. Without observing the full network, the study cannot measure whether the successful chains actually followed near-optimal paths or just happened to complete.

Speaker notes

The chain-death problem is the deepest limitation: the experiment conflates the *ability* to find short paths with the *willingness* to search. Milgram's original experiment had the same issue. No routing experiment can fully separate topology from motivation without observing the complete network and making forwarding mandatory — which is ethically impossible. This is the navigational gap in its purest empirical form.

Takeaway

Dodds et al. (2003) confirmed Milgram at scale: short paths exist and people can sometimes find them. But the 1.6% completion rate reveals that *motivation*, not topology, is the primary bottleneck. The navigational gap is real — short paths exist but are rarely found in practice.

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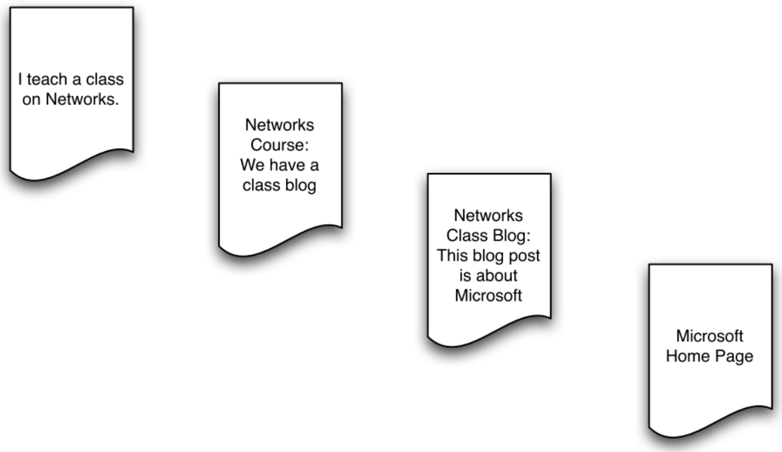
The Web as a Directed Small World

Guiding Question: What happens to navigability when links are directed?

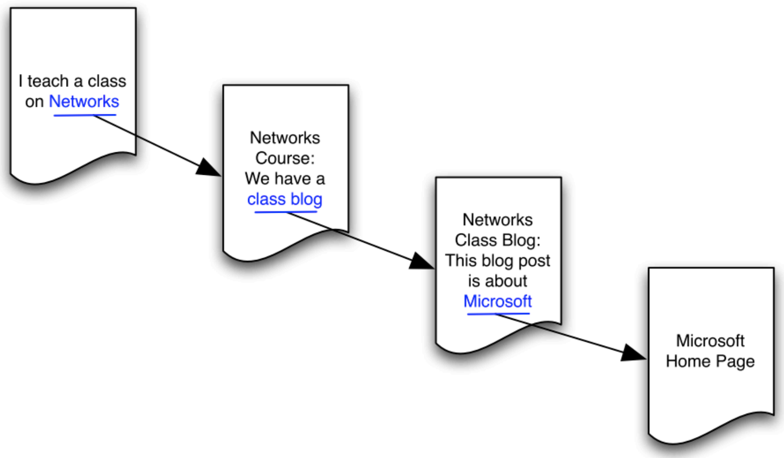
Speaker notes

The Web is the largest information network ever built, and its links are directed — a page can link to another without being linked back. This breaks the symmetry that underpins both Watts-Strogatz and Kleinberg: in a directed graph, reachability is not mutual, and the "small world" may only exist in part of the network.

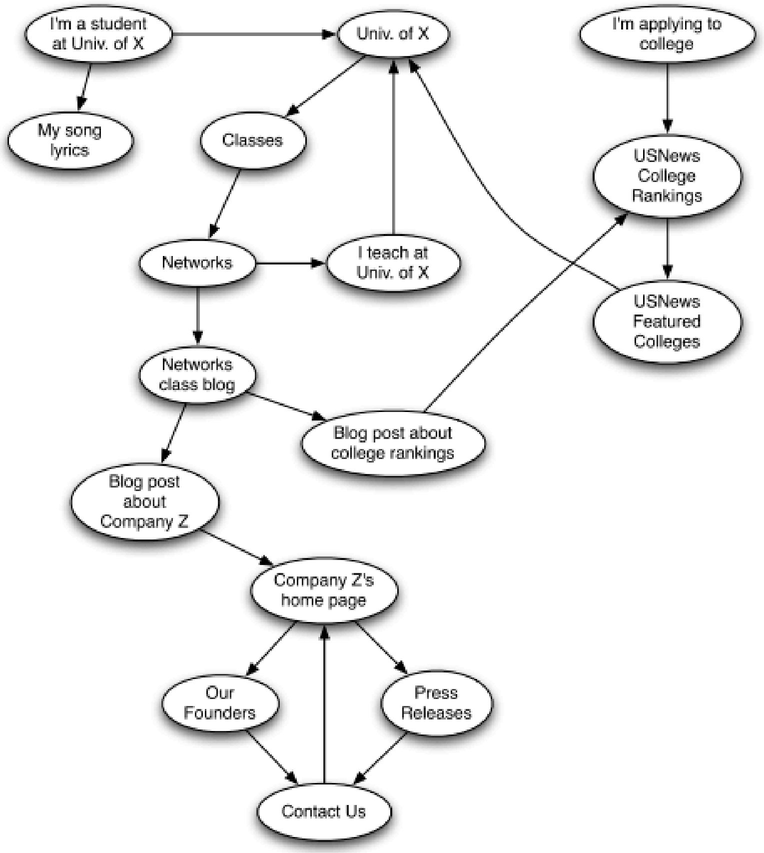
From Social to Information Networks



Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010

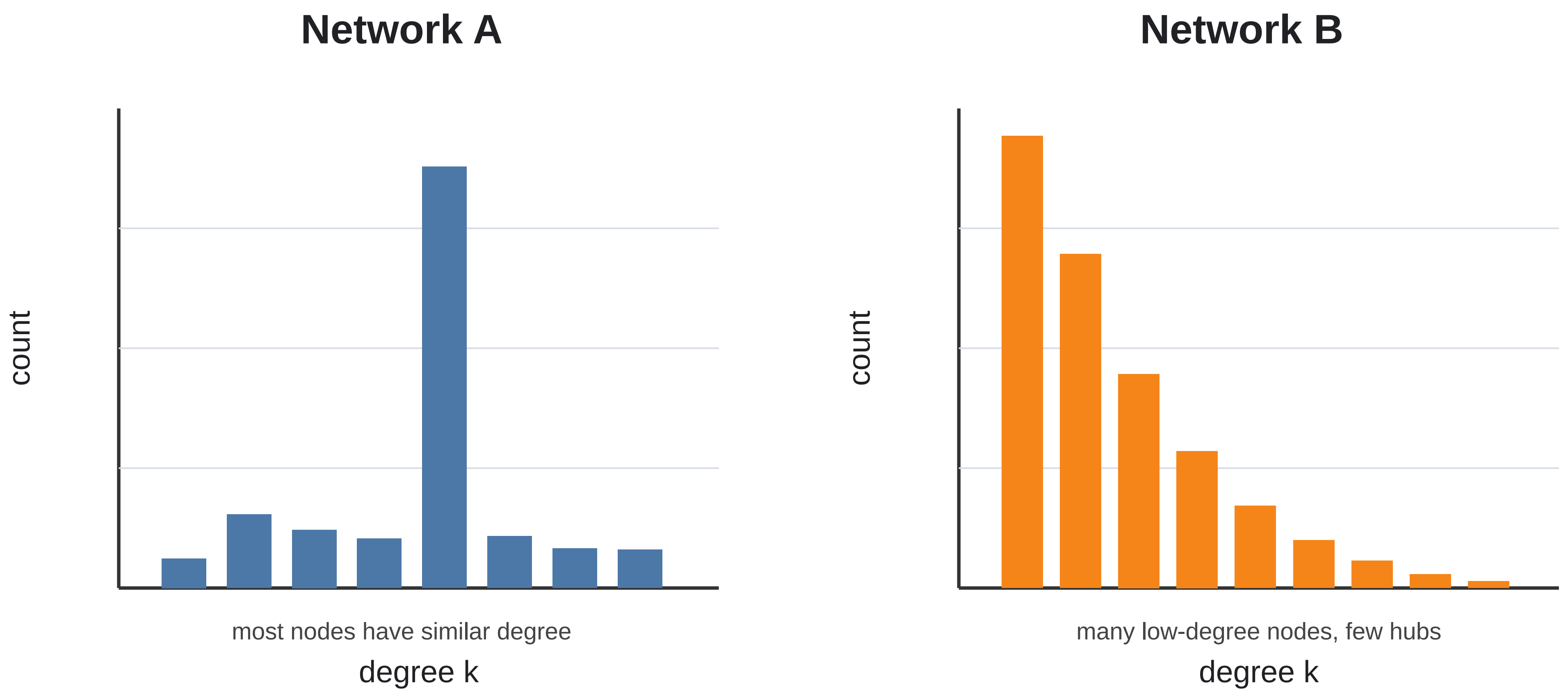


Source: Easley & Kleinberg 2010

Hyperlinks are **directed** — page A can link to page B without B linking back. This means reachability is asymmetric: you can follow links *from* A *to* B but not necessarily in reverse.

Scale-Free Structure of the Web

The Web is not degree-homogeneous. Most pages receive few incoming links, while a small number of pages receive enormous numbers of links.



Scale-free-like Web. A directed information network whose in-degree distribution is heavy-tailed / approximately power-law:

$$P(k_{in}) \propto k_{in}^{-\alpha}$$

There is no single "typical" page: attention concentrates around hubs (Barab'asi & Albert 1999).

Speaker notes

This is the better home for scale-free graphs than the opening small-world module. On the Web, degree splits into out-degree and in-degree. Out-degree is how many pages a page links to; in-degree is how many pages link to it. The famous scale-free claim is mainly about in-degree: most pages receive few links, while a small number of pages become authorities or hubs with enormous visibility. This also connects directly to PageRank from L04: incoming links are treated as endorsements, so the heavy-tailed in-degree distribution becomes a heavy-tailed attention distribution.

Barabási-Albert: A Model of Web Growth

Preferential attachment. New pages tend to link to pages that are already visible.

In a simplified directed Barabási-Albert model:

1. Start with a small connected Web.
2. Add one new page at a time.
3. Each new page creates m outgoing hyperlinks.
4. Existing page v receives a new link with probability proportional to its current in-degree:

$$P(\text{new page links to } v) \propto k_v^{\text{in}}$$

Speaker notes

Use this as a stylized model, not a complete theory of the Web. In the original Barabási-Albert model, attachment probability is proportional to degree. For the Web, the directed interpretation is in-degree: pages with many incoming links are easier to discover, cite, copy, rank, and reuse, so they tend to receive even more incoming links. In practical models one often adds baseline attractiveness so brand-new or low-degree pages can still receive links, but the core idea remains preferential attachment.

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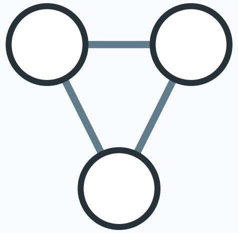
This is the **rich-get-richer** mechanism: popular pages attract more links, which makes them even more visible.

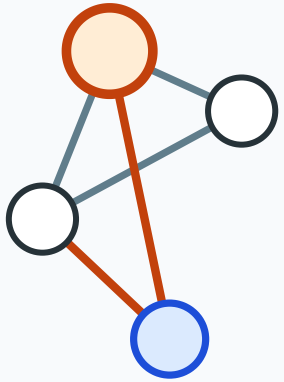
Barabási-Albert in Four Steps

1. Seed



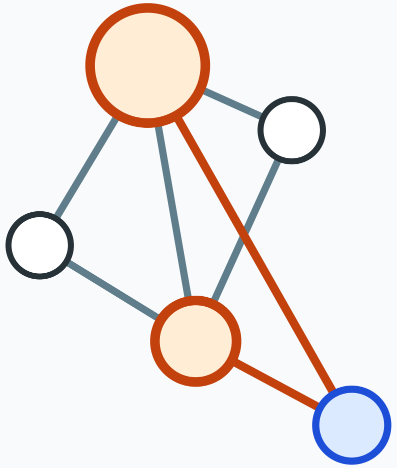
start with a small
connected graph

2. Grow



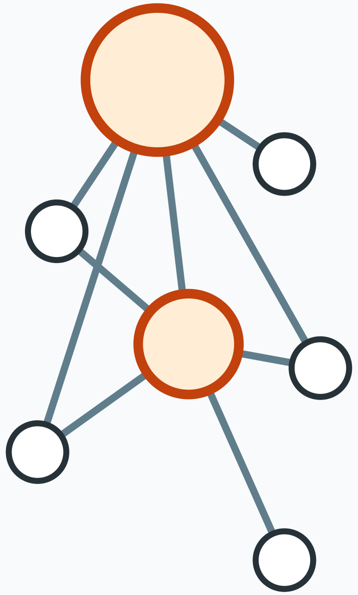
add one node
and m edges

3. Prefer hubs



choose endpoints with
probability proportional to degree

4. Hubs emerge



rich-get-richer growth
creates a heavy tail

Preferential attachment rule: probability of attaching to node v is proportional to its current degree k_v.

Nodes that are already well connected become even more likely to receive new edges.

Speaker notes

Use this as the visual analogue to the Watts-Strogatz four-step slide, but frame it as a Web-growth model. The graph is not rewired from a fixed lattice; it grows. Each new page chooses targets with probability proportional to current popularity. Over many steps this produces a heavy-tailed distribution: many obscure pages and a few highly linked authorities.

What Preferential Attachment Explains

Web pattern	Preferential-attachment explanation	Limitation
Heavy-tailed in-degree	Popular pages attract more links	Not every popular page is high quality
Short paths to authorities	Hubs connect many topics	Direction still restricts reachability
Attention inequality	Visibility compounds over time	Aging and novelty also matter
Link copying	Authors imitate existing link lists	Communities and spam distort the signal

Speaker notes

This slide narrows the claim. Preferential attachment is useful because it explains a robust macro-pattern with a simple local mechanism. But the Web is not just a pure rich-get-richer process. Content quality matters; new pages can become popular; old pages decay; search engines amplify some links and hide others; topic communities create dense regions; spam and SEO manipulate link structure; and many links are copied from templates, blogrolls, documentation, and navigation menus.

What Preferential Attachment Explains

Web pattern	Preferential-attachment explanation	Limitation
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Link copying	Authors imitate existing link lists	Communities and spam distort the signal

> BA explains why hubs can emerge, but it is not a full model of the Web. The real Web also depends on content quality, topical communities, search engines, aging, copied links, platforms, and incentives.

Kleinberg on the Web: Structured Shortcuts

Scale-free hubs help explain **which pages become highly linked**. Kleinberg-style models explain a different question: **when can local navigation make progress toward a target?**

Model idea	Web interpretation
Preferential attachment	popular pages attract more incoming links
Kleinberg long-range links	hyperlinks connect across topical or semantic distance
Greedy routing	users follow links that seem closer to the information goal

Speaker notes

This connects the Web module back to the navigability theorem without claiming that the real Web literally satisfies Kleinberg's grid assumptions. For Web navigation, "distance" is not geographic distance; it may be topical, semantic, institutional, linguistic, or social. A hyperlink from a local topic page to a related global authority is a structured shortcut. If links reflect useful semantic distances, local browsing can make progress. But scale-free degree alone does not guarantee navigability: a hub can be popular without being semantically helpful for a particular target.

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Model idea	Web interpretation
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Greedy routing	users follow links that seem closer to the information goal

> Kleinberg does **not** imply scale-free. It uses a power law over **distance**, not over **degree**. The Web can be scale-free-like and small-world-like at the same time, but those are separate properties.

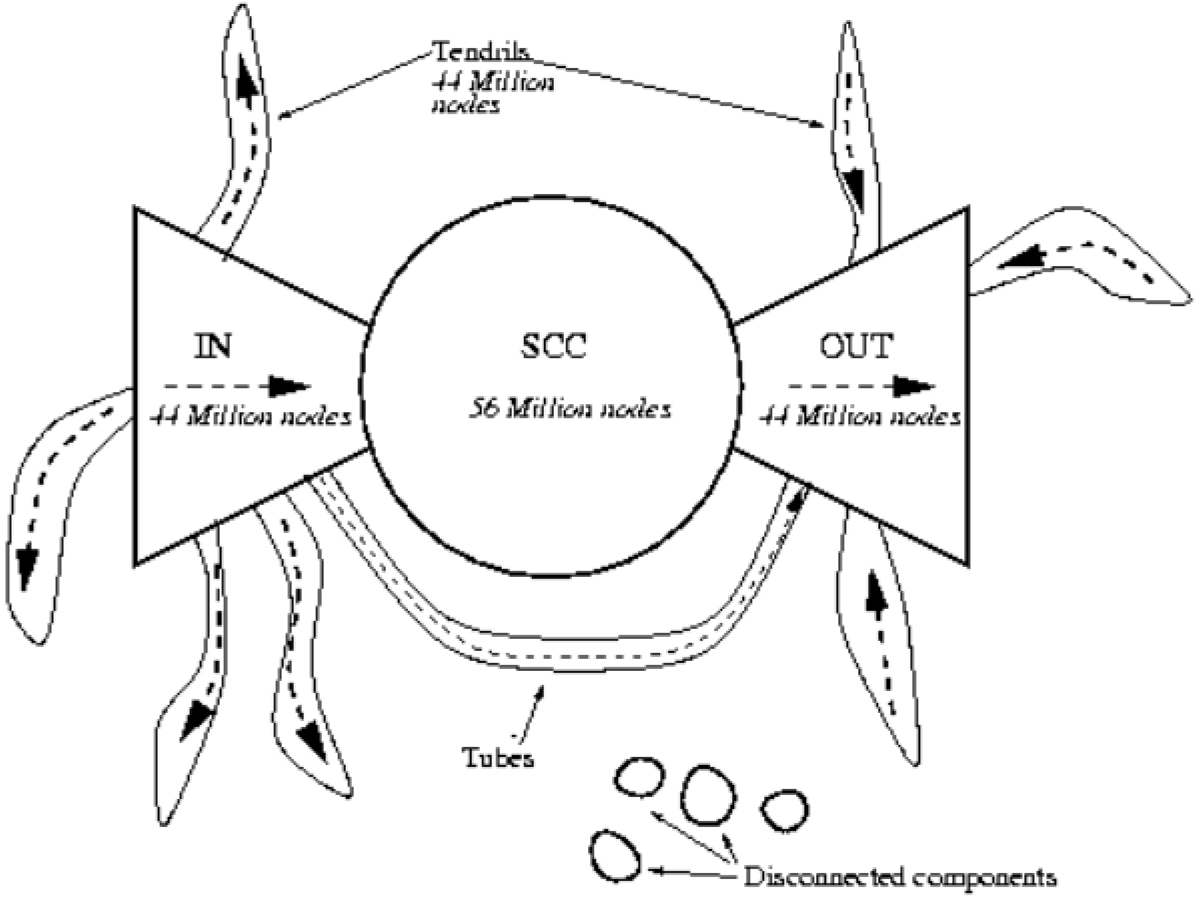
Web Synthesis

The Web can be seen as both **scale-free-like** and **small-world-like**. Preferential attachment explains the emergence of hubs, because popular pages attract more links over time. Kleinberg-style models explain navigability through structured shortcuts, especially when links reflect topical or semantic distance. However, the real Web is shaped by additional mechanisms such as content quality, topic communities, search engines, aging, and copied links.

Speaker notes

This is the synthesis slide for the Web module. It states the balanced conclusion: BA gives a plausible mechanism for heavy-tailed in-degree; small-world structure explains short directed paths within the reachable core; Kleinberg gives the condition for local navigability; and real Web formation is richer than any one model. This is also the right place to remind students that models are explanatory lenses, not complete replicas.

The Bow-Tie Structure



Source: Easley & Kleinberg 2010, p. Ch. 13

Broder et al. (2000) crawled ~200 million web pages and found the Web decomposes into:

- **SCC (Strongly Connected Component):** ~28% of pages — every page reachable from every other via directed paths. *This* is the "small world" of the Web.
- **IN:** pages that can reach the SCC but are not reachable from it.
- **OUT:** pages reachable from the SCC but that cannot reach back.
- **Tendrils and tubes:** pages connected to IN or OUT but not to the SCC.

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Speaker notes

The bow-tie decomposition is the directed analogue of connected components. The SCC is the only part where mutual reachability holds — where "six degrees" makes sense. The IN, OUT, and tendril components break the small-world property because paths exist in one direction but not the other. Roughly 72% of the Web is *not* in the SCC, meaning the "small world" is a minority of the full graph.

Navigability in Directed Networks

The small-world property and navigability both change in directed networks:

- **Short paths exist** within the SCC — the strongly connected core is a small world (~16 clicks between random page pairs within SCC).
- **Short paths do not exist** from OUT back to SCC, or between tendrils.
- **Greedy search is harder** because you can only follow outgoing links — you cannot "pull" information from a page that doesn't link to you.

Speaker notes

This connects to L04: PageRank (eigenvector centrality on the directed Web graph) is the tool that makes centralized Web search possible. The Web is simultaneously a small world (within the SCC) and not navigable by local greedy search (because of directionality). The resolution — a centralized search engine — sidesteps the navigational gap rather than solving it.

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Search engines (Google, 1998) solve the navigability problem by building a *global index* — they observe the entire graph and compute optimal paths. This is the opposite of Kleinberg's decentralized search: it works precisely because it is *centralized*.

Takeaway

The Web's directed links create asymmetric reachability. Only the Strongly Connected Component (~28%) is a "small world." Search engines solve the navigability problem through centralized indexing — the opposite of Milgram-style decentralized routing.

Quick Check

A directed network has 1000 nodes. The SCC contains 300 nodes. The IN component has 200 nodes, and the OUT component has 250 nodes. The rest are tendrils.

1. From a random node in IN, can you reach a random node in OUT? Explain your path.
2. From a random node in OUT, can you reach a node in IN?
3. What fraction of all possible source-target pairs have a directed path between them?

Speaker notes

Give them 4 minutes. (1) Yes: IN → SCC → OUT. The IN component can reach the SCC by definition, and the SCC can reach OUT by definition. So IN → SCC → OUT is a valid directed path. (2) No — OUT cannot reach back to the SCC (by definition), and from the SCC you can reach OUT but not IN. OUT → IN has no path. (3) This is harder. Pairs within SCC: 300×299. IN→SCC: 200×300. IN→OUT via SCC: 200×250. SCC→OUT: 300×250. Others require more analysis. The point is that far fewer than 50% of all pairs are mutually reachable.

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A directed network has 1000 nodes. The SCC contains 300 nodes. The IN component has 200 nodes, and the OUT component has 250 nodes. The rest are tendrils.

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Quick Check — Solution

1. Yes: IN → SCC → OUT. By definition, IN nodes can reach SCC, and SCC nodes can reach OUT. The path goes through the SCC as a relay.

2. No. OUT nodes cannot reach back to the SCC (that is what defines them as OUT, not SCC). So OUT → IN has no directed path.

3. Reachable pairs include: SCC→SCC (300×299), IN→SCC (200×300), SCC→OUT (300×250), IN→OUT via SCC (200×250). That totals ~$240,000$ pairs out of $1000 \times 999 \approx 10^6$ — roughly **24%**. Most pairs are *not* mutually reachable.

Wrap-Up — Lecture 07

Question	Key idea	Takeaway
Why are large networks close?	Logarithmic distances	Short paths can exist even in huge networks.
How can networks be clustered *and* short?	Watts-Strogatz shortcuts	A few long-range edges collapse L while C stays high.
Can local actors find short paths?	Kleinberg's theorem	Findability needs the right multi-scale link geometry.
Do people actually complete searches?	Milgram; Dodds et al.	Many paths die because motivation and information are limited.
What changes on the Web?	Directed bow-tie structure	Reachability is asymmetric; only the SCC behaves like a small world.
Why is the Web hub-dominated?	Barabási-Albert	Preferential attachment can produce scale-free-like in-degree.
Is the Web locally navigable?	Structured shortcuts	Topical/semantic links help, but search engines centralize navigation.

> Lecture 07 separates **existence** from **findability**: short paths may be present in the graph, but local actors can use them only when the network provides navigable structure.

Looking Ahead: Lecture 08

How do things *spread* on networks — diseases, information, behaviors? The structure we have built (ties, communities, balance, small-world shortcuts) now becomes the substrate for dynamic processes: contagion, diffusion, and cascading failure.

Reading

Chapter 20 of Easley & Kleinberg (2010) covers the small-world phenomenon and Watts-Strogatz. Barabási & Albert (1999) introduces scale-free networks and power-law degree distributions. Kleinberg (2000), *The Small-World Phenomenon: An Algorithmic Perspective*, is the source for the decentralized search theorem. Dodds, Muhamad & Watts (2003) is the email experiment. Broder et al. (2000) is the bow-tie structure of the Web.

Speaker notes

Close the lecture by distinguishing the three layers one more time: existence of short paths (small-world property), structural explanation (Watts-Strogatz), and decentralized findability (Kleinberg plus empirical search experiments). Lecture 08 introduces the final conceptual shift: from structure to dynamics. The graph is no longer just an object to analyze — it becomes a *medium* through which processes propagate.

Image Credits & References

Barab'asi, Albert-L'aszl'o and Albert, R'eka (1999). Emergence of Scaling in Random Networks. *Science*.

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning about a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/